

BTEC Level 3 Applied Science · Unit 1 Physics · Worksheet

Wave Properties

Original JDScience practice. These questions were written for this resource and are not official exam-board items.

A Retrieval

1. Define a transverse wave and a longitudinal wave.
2. Give one example of each.
3. State the principle of superposition.

B Knowledge

4. What is a compression and what is a rarefaction?
5. When are two points on a wave in phase?
6. Why can light be polarised but sound in air cannot?

C Application

7. Classify ultrasound, microwaves and stadium sound as transverse or longitudinal.
8. Two crests are 6.0 cm apart and $\lambda = 2.0$ cm. Are they in phase? Explain.
9. Describe what happens when a crest meets a trough of equal amplitude.

D Exam-style practice

10. Describe the difference between transverse and longitudinal waves, with an example of each. [4]
11. State the principle of superposition. [2]
12. Explain why sound from a loudspeaker cannot be polarised. [2]

E Challenge

13. A student says a sine-wave drawing of sound proves sound is transverse. Write a correction a teacher could use.